

TASK 1 DOCUMENTATION

Embedded Web Server using STM32F767ZI

1. Aim

To configure the **STM32F767ZI** microcontroller as an embedded web server using Ethernet and control LEDs via a web browser.

2. Required Components

- STM32F767ZI Nucleo Board
- Ethernet Cable (RJ45)
- Router
- Laptop/PC

3. Software Requirements

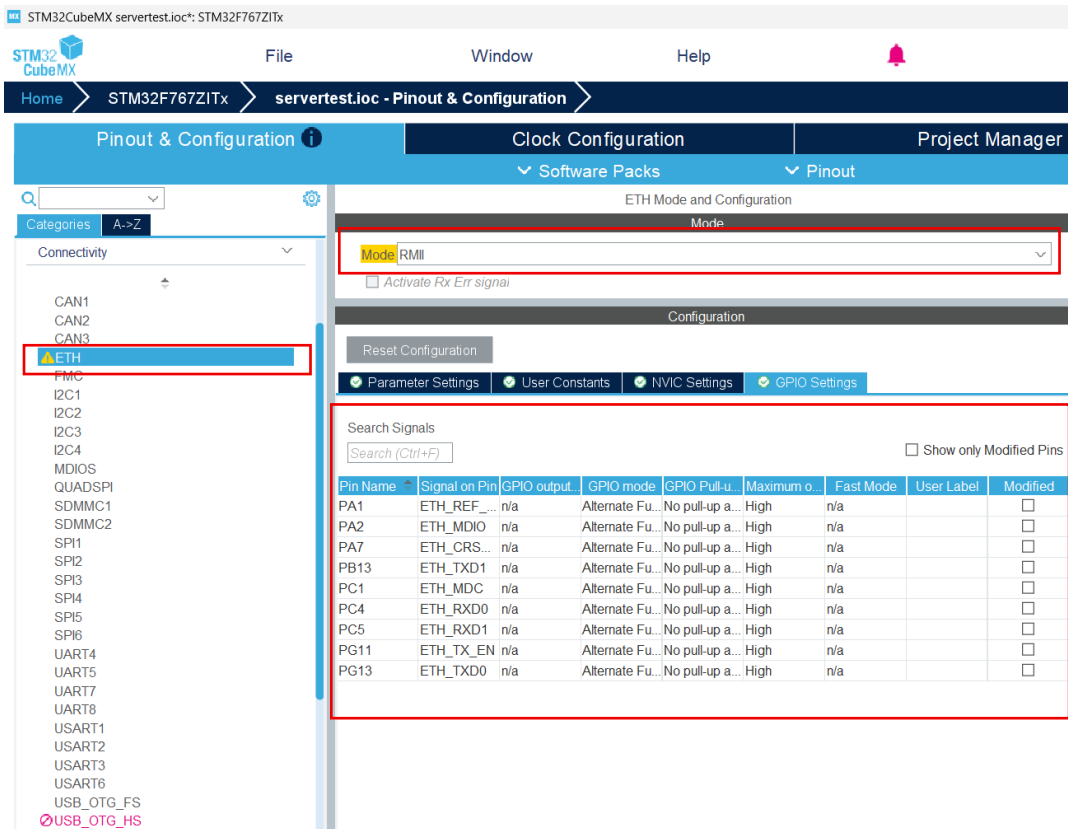
- STM32CubeIDE
- STM32CubeMX
- Perl (for fsdata generation)

4. Step-by-Step Procedure

Step 1: Create New Project

1. Open STM32CubeIDE
2. Click File → New STM32 Project
3. Select board: STM32F767ZI Nucleo
4. Click Finish

Step 2: Enable Ethernet (ETH)

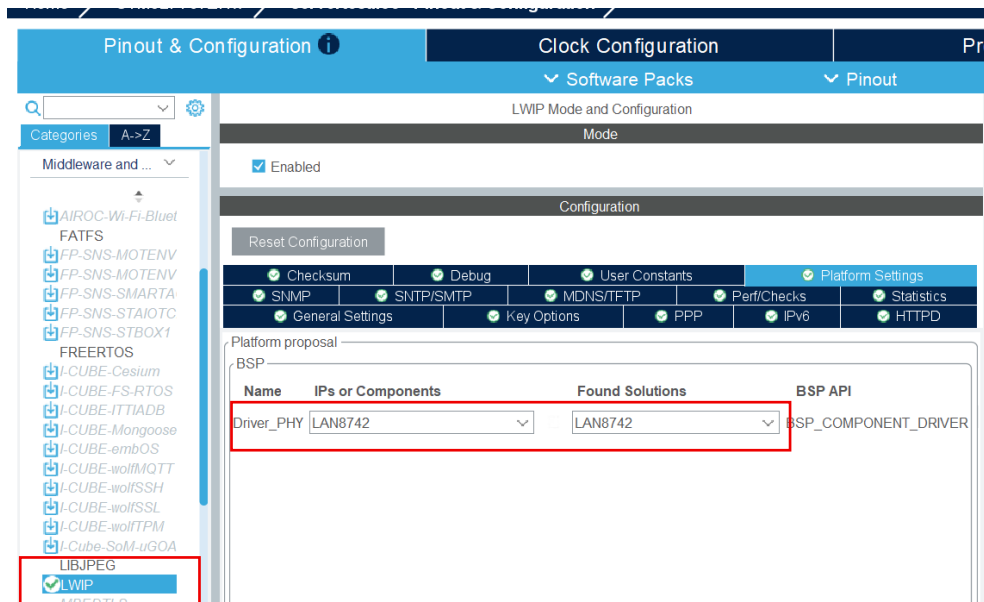


1. Go to Connectivity → ETH
2. Set:
 - Mode = RMII

Step 3: Configure Ethernet Pins

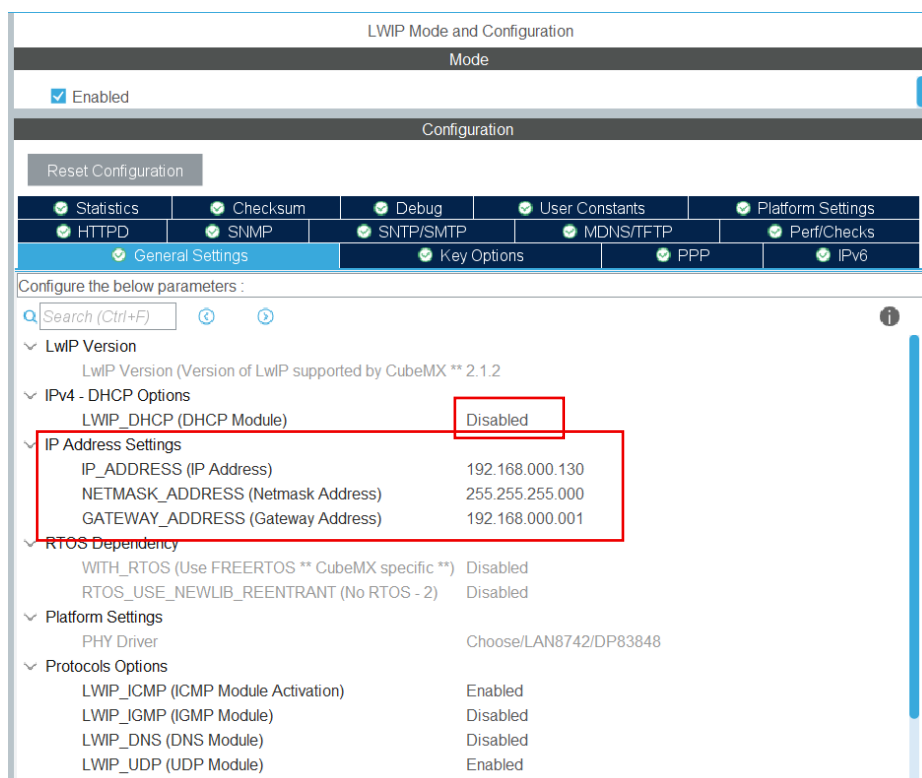
1. Ensure following pins are enabled:

Pin	Function
PA1	REF_CLK
PA2	MDIO
PC1	MDC
PA7	CRS_DV
PC4	RXD0
PC5	RXD1



Step 5: Configure IP Address

1. Go to LwIP → General Settings
2. Disable DHCP
3. Set:



Step 6: Enable HTTP Server

1. Go to **LwIP** → **HTTPD**
2. Enable:
 - HTTPD Server

Step 7: Configure GPIO for LEDs

1. Go to **Pinout**
2. Enable:
 - PB0 → Output (Green LED)
 - PB7 → Output (Blue LED)
 - PB14 → Output (Red LED)

Step 8: Configure Clock

1. Go to Clock Configuration
2. Click Resolve Clock Issues Automatically
3. Ensure system clock ≈ 216 MHz

Step 9: Generate Code

1. Click **Save (.ioc file)**
2. Click **Generate Code**

Step 10: Modify main.c

► Add include:

```
#include "lwip/apps/httpd.h"
```

► Initialize HTTP server:

```
httpd_init();  
http_set_cgi_handlers(&LED_CGI, 1);
```

► Add inside while loop:

```
MX_LWIP_Process();

HAL_GPIO_WritePin(GPIOB, GPIO_PIN_0, GreenLED);
HAL_GPIO_WritePin(GPIOB, GPIO_PIN_7, BlueLED);
HAL_GPIO_WritePin(GPIOB, GPIO_PIN_14, RedLED);
```

► CGI Handler (LED Control)

This function processes URL requests:

```
const char *LedControlCgiHandler(...)
```

Reads:

- green=ON/OFF
 - blue=ON/OFF
 - red=ON/OFF
- Controls LEDs accordingly

Step 11: Generate Web Files (fsdata)

1. Create HTML files:

- index.html
- 404.html

2. Run:

```
perl makefsdata.pl
```

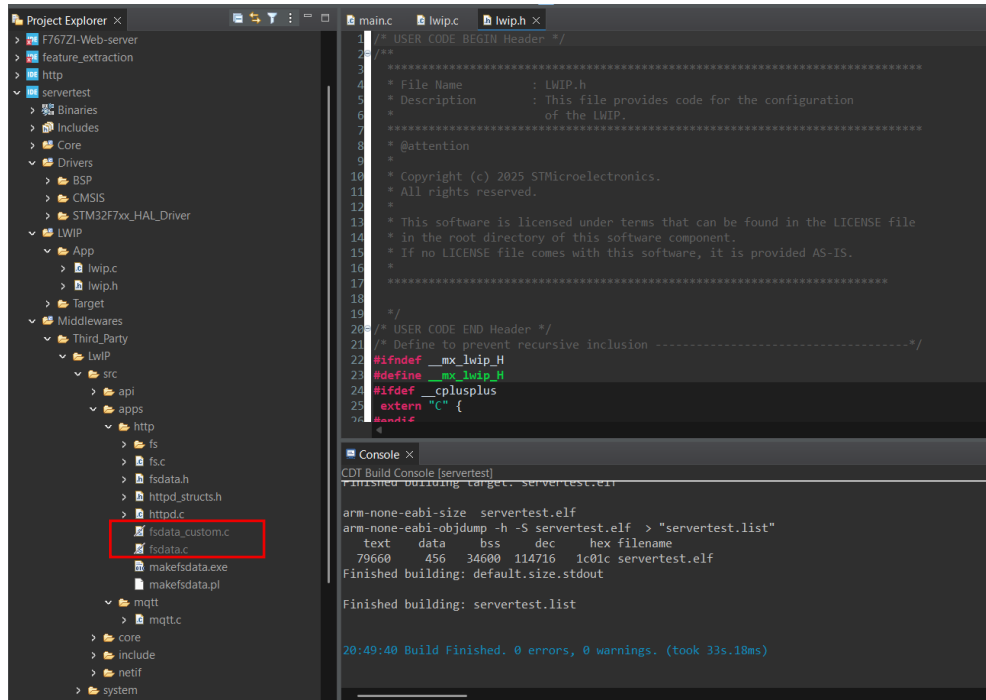
3. Generated files:

- fsdata.c
- fsdata_custom.c

Step 12: Add Files to Project

Path:

Middleware → Third_Party → LwIP → src → apps → http



Step 13: Exclude fsdata Files from Build

1. Right click files
2. Select:
 - Resource Configuration → Exclude from Build

Step 14: Build Project

- Click **Build**
- Ensure:
 - No errors

Step 15: Hardware Setup

1. Connect Ethernet cable:
 - STM32 → Router
2. Connect PC to same network

Step 16: Flash and Run

1. Flash code to STM32
2. Restart board
3. Observe LED blinking

Step 17: Test Web Server

Open browser:

<http://192.168.0.130>

NOTES

- IP must match router network
- Same WiFi/network required
- Correct PHY (LAN8742) is critical
- fsdata files must be generated properly